

PH3202 (Intermediate EM)

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Class Test 1

Question 1 (5 marks)

Find the electric field at a distance r away from the midpoint of a straight line segment of length L , which carries a line charge density λ . For the case where $r \ll L$, express the electric field at the point r . Does $\vec{E}$ depend on length L ?

Question 2 (5 marks)

Find the solution of the Laplace's equation ($\nabla^2 V = 0$) in 1D. Show that the potential $V(x)$ at any point can be expressed as the average of its left and right values of the potential. Does the potential admit any local extremum?